I've generated the dataset and run the analysis, and the findings look good. Now I'm checking the charts and then adding the SQL part, README and career guide.I built a complete data analytics portfolio project and ran it end to end, so every number in the README comes from real output. The project is **Retail Sales, Profit & Customer Retention Analysis**, using Python, SQL and a dashboard guide.

The data is simulated. It's an online store with 3,000 customers and about 6,000 orders over 2023–24, deliberately messy (duplicate rows, inconsistent city names, missing values). Say "simulated" in your README and in interviews. The career guide explains how to repeat the analysis on a real dataset (Olist or Superstore from Kaggle), which makes the project much stronger.

What it finds

- Electronics is 73% of revenue but only an 8% margin.
- Discounts of 30% lose money (margin −16% overall, −21% on Electronics).
- Referral and Email customers come back (75% and 72% repeat), but Social Media customers mostly don't (35%).
- Champions are 7% of customers but 19% of revenue, and 1,234 repeat customers have gone quiet.
- Late deliveries pull ratings from 4.4 down to 2.7 stars.

The README turns these into four business recommendations.

What's inside the zip

- `data/generate_data.py` creates the four CSV files.
- `analysis.py` does the cleaning, KPIs, seasonality, categories, discounts, channels, RFM segments, cohort retention and delivery analysis. It writes 7 charts and summary tables. You can run it like a notebook in VS Code.
- `sql/` and `run_sql.py` hold 9 SQL queries (CTEs, joins, `LAG`, `RANK`). They give the same answers as the Python.
- `README.md` is the portfolio write-up, ready to push to GitHub.
- `docs/CAREER_GUIDE.md` has the Power BI/Tableau dashboard steps, resume bullets, interview questions and a 7-day plan.

What you still need to do

1. Build the dashboard yourself in Power BI or Tableau Public. I couldn't make a `.pbix` file, but the guide gives you the layout and measures. Recruiters click on this first.
2. Learn the project well enough to explain it without notes. Interviewers will ask why you made each choice.
3. Add a real-data version. I didn't verify any Kaggle dataset from here, so check that Olist or Superstore loads before you plan around it.

The monthly revenue chart shows no clean festive peak, because a few large electronics orders swing each month. I said so in the README rather than force a seasonality story.

I wrote `PROJECT_EXPLAINED.md`, and it's also inside the updated zip under `docs/`.

What's in it

1. **Pitch and walkthrough:** a 30-second pitch and a 2-minute walkthrough script to memorise.
2. **Step by step:** all 13 steps of the project, each with what we did, why it matters and how it works in the code.
3. **Numbers cheat sheet:** the key figures to quote.
4. **Plain-words concepts:** margin vs revenue, RFM, cohorts, CTEs and window functions.
5. **Interview Q&A:** 16 likely questions with model answers, including "Is this real data?" and "What are the limitations?"
6. **How to run:** setup for Windows and Mac, the exact commands, what each script produces, a troubleshooting table and the GitHub push steps.
7. **Checklist:** a final checklist to tick off before the interview.

I tested the code on Python 3.12 with pandas 3.0, but not on Windows, so use the troubleshooting table if something fails there. If your numbers differ slightly after re-running, quote what `outputs/findings.json` shows.

Answer question 2 ("Did you use AI tools?") honestly. Interviewers often ask, and they will probe the details. Before you apply, run every script, change one rule such as the discount cap or a segment threshold, and make sure you can explain every line.

I can also convert this document to Word or PDF if you'd rather study it that way.